

# AI/ML Engineer L1 Interview Questions & Answers (Azure + RAG + Python)

## 1. Why Azure?

Managed AI services, security, scalability, Azure OpenAI, AI Search, Blob Storage, Key Vault, Monitor.

## 2. Why Azure is private?

Private Endpoints, VNets, RBAC, Managed Identity, encryption protect enterprise data.

## 3. Input data

PDF, CSV, DOCX, TXT, HTML, JSON. Files in Blob, embeddings in vector DB.

## 4. Model selection

Choose GPT-4/GPT-5 based on accuracy, latency, cost, reasoning and context window.

## 5. Monitor model

Latency, token usage, cost, quality, hallucinations, user feedback, Azure Monitor, LangSmith.

## 6. Validate model

Use RAGAS, human evaluation, A/B tests and user feedback to improve prompts/retrieval/model.

## 7. Without LLM

Ingestion and retrieval still work; chatbot cannot generate natural-language answers.

## 8-9. 10 GB CSV

Process in streaming chunks, clean, transform, embed/store each chunk, then discard from memory.

## 10. When not DL

Small datasets, explainability needs, simple problems, limited compute.

## 11. Precision vs Recall

Precision:  $TP/(TP+FP)$ . Recall:  $TP/(TP+FN)$ .

## 12. LLM internals

Tokenization → Embeddings → Transformer → Attention → Next-token prediction.

## 13. Set vs Dict

Set: unique values. Dict: key-value pairs.

## 14. Mutable/Immutable

Mutable: list, dict, set. Immutable: tuple, string, int.

## **15. Iterator/Generator/Decorator**

Iteration protocol, lazy yield, function wrapper.

## **16. Thread**

Concurrent execution, ideal for I/O-bound tasks.

## **17. PyTorch**

Tensors, nn.Module, DataLoader, optimizers, autograd.

## **18. RAG**

Chunk → Embed → Vector DB → Retrieve → (Optional rerank) → LLM → Answer.

## **19. Reranking**

Optional but improves retrieval quality before the LLM generates the final response.

## **20. Chunk size**

Usually 300–500 tokens with 50–100 overlap; validate using RAGAS and retrieval metrics.